

EARTH VISION-X

Explainable Foundation AI Satellite Change Detection Report

Primary Change Detected:	Urban Expansion
Confidence Score:	11.4%
Affected Area:	6.55 sq km (100.0%)
Inference Time:	0.669 seconds
Timestamp:	2026-07-30 18:49:17 UTC

AI Executive Summary & Explainability Insights

Between 2021 and 2024, approximately 100.0% (6.55 sq km) of spatial surface area underwent significant environmental transformation dominated by **Urban Expansion**. The foundation AI model (Swin Transformer Siamese) evaluated this change event with a confidence score of **11.4%**. Attention rollout maps indicate that the Vision Transformer self-attention layers focused primarily on boundary gradients, canopy loss frontiers, and spectral index variance. Secondary urban infrastructure development increased by 8.0% across the surrounding buffer zone.